

Welfare Analysis of a Forest-Residue Sustainable Aviation Fuel Supply Chain under Endogenous Demand and Price Uncertainty

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Abstract

We develop a county-aggregated two-stage stochastic mixed-integer program for a southeastern U.S. forest-residue-to-sustainable-aviation-fuel (SAF) supply chain, embedded in a partial-equilibrium welfare framework with endogenous SAF demand. First-stage variables locate and size depots and biorefineries; the second stage allocates flows under SAF price scenarios drawn from a kernel-density estimate of historical wholesale prices. SAF is treated as a buyer-segmented market driven by ESG offtake commitments and federal/state mandates, with linear inverse demand calibrated to the U.S. Department of Energy’s 3 billion-gallon-per-year SAF Grand Challenge target. To bound the recourse-problem expected cost, we use a candidate-evaluation procedure based on parametric LP-only sweeps over wait-and-see solutions; the same machinery yields a Pareto frontier on the (expected loss, $\text{CVaR}_{0.05}$) plane.

The chain-versus-import break-even SAF price is \$5.83/gal and the topological-expansion threshold is \$6.05/gal. The candidate-evaluation upper bound on the recourse expected cost is \$0.925 B/yr. The endogenous-demand welfare equilibrium gives \$17.27 B/yr at $\varepsilon_{\text{SAF}} = -0.5$, with a robustly negative producer surplus implying a per-gallon production-credit gap of \$0.21–\$1.48/gal across the elasticity grid — comparable in magnitude to the IRA Section 45Z credit. The 20-candidate Pareto frontier has two undominated plans, with an implied risk premium of \$7.1 M/yr; the deterministic equilibrium plan is dominated, with a \$300 M/yr stochastic-vs-deterministic gap on the worst-5% tail.

Keywords: sustainable aviation fuel; supply chain; stochastic programming; consumer surplus; risk-aware optimization; CVaR ; expected value of perfect information.

JEL codes: Q42 (Alternative Energy Sources), Q56 (Environment and Development), C61 (Optimization), D61 (Welfare).

1 Introduction

1.1 Motivation

The U.S. SAF Grand Challenge sets a production goal of 3 billion gallons per year by 2030 (DOE, 2021). At present, U.S. SAF production stands at roughly 0.5–0.7 B gal/yr (EIA, 2025) — about one-fifth of the goal — at a wholesale price of approximately \$9/gal, which incorporates a \$5.68/gal

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premium over fossil Jet-A (IATA, 2025). Closing this six-fold gap will require a coordinated buildout of pretreatment depots and biorefineries, with feedstock decisions embedded in a market that is itself transitioning. The southeastern United States holds the largest commercially recoverable supply of forest harvest residue in the country (Roni et al., 2017), making it a leading candidate region for residue-based Fischer–Tropsch (FT) and alcohol-to-jet (ATJ) capacity at scale. Our prior work establishes the deterministic supply-chain structure for this region under alternative preprocessing-and-conversion stacks (Li et al., 2024) and assesses feedstock ash-content variability under a fast-pyrolysis/hydrotreatment pathway (Li et al., 2025); the present paper extends that line of work with endogenous SAF demand, a welfare framework, and risk-aware planning under price uncertainty.

1.2 The economic problem

Existing supply-chain optimization models for residue-to-SAF treat the problem as cost minimization under a fixed selling price (Sharma et al., 2013; Lin et al., 2014; Roni et al., 2017). This framing has three weaknesses from an economic perspective. An exogenous SAF price assumes away market clearing. Fixed-price formulations cannot decompose project value into consumer surplus, producer surplus, and externality components — the decomposition needed to evaluate policy counterfactuals (Pigou, 1920; Weitzman, 1974; Pizer, 2002). And deterministic NPV models offer no risk metric, although investment decisions in an emerging fuel market are dominated by payoff variance rather than mean (Dixit and Pindyck, 1994; Rockafellar and Uryasev, 2000).

1.3 Contributions

At the heart of the model is an endogenous SAF demand curve, $P_{\text{SAF}}(Q) = a - bQ$, calibrated to corporate offtake commitments and the SAF Grand Challenge target. We treat SAF as buyer-segmented — driven by ESG mandates, federal and state credits, and voluntary procurement — rather than as a perfect substitute for fossil Jet-A. The distinction is consequential: under the perfect-substitute alternative, a social planner with $MC_{\text{SAF}} > P_{\text{fossil}}$ never builds the chain, leaving the welfare exercise empty. The buyer-segmented demand curve lets the price clear at the SAF-specific willingness to pay and supports a Marshallian welfare decomposition into consumer surplus, producer surplus, and externality terms (§3.3).

Computing the recourse-problem expected cost is the central methodological obstacle. The extensive-form deterministic equivalent fails to converge at our problem scale, and integer Benders, while feasible, is expensive enough to be unattractive as a routine tool. We bound $\mathbb{E}[\text{PVC}_{\text{RP}}^*]$ from above by a parametric LP sweep over wait-and-see first-stage solutions: each wait-and-see plan is fixed in turn as a non-anticipative candidate and re-evaluated under all KDE scenarios; the minimum candidate-conditional expected cost is the constructive upper bound. The full sweep finishes in 18.8 minutes versus the failed extensive form, with a headline value of \$0.925 B/yr (\$5.5). The procedure is portable to other large-scale recourse problems where the extensive form is borderline tractable.

For risk-aware planning we adopt the Rockafellar–Uryasev CVaR functional but apply it to welfare loss, $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$, rather than to project cost. Welfare loss is the decision-relevant scalar for a social planner, and the policy parameter λ then has a direct interpretation as aversion to worst-tail welfare loss. Rather than solving the joint MILP — which inherits the recourse-problem scale issues — we trace a discrete Pareto frontier on the $(\mathbb{E}[L], \text{CVaR}_\alpha(L))$ plane by enumerating all 20 wait-and-see candidates. Two are undominated, with an implied risk premium of \$7.1 M/yr; the deterministic

Path B equilibrium plan is itself dominated, with a \$300 M/yr stochastic-vs-deterministic gap on the worst-5% tail (§5.6).

A methodological refinement underwrites all of the above: a `shapely`-based hex-to-county area-share aggregation, exact under spatial uniformity. Of 14,040 hexes in the reference grid, 8,864 (63%) straddle county boundaries, so single-FIPS assignment introduces a measurable bias that the area-share method removes. The aggregated MILP matches the hex-level reference within 0.15% on cost and 0.07% on residue, while shrinking location nodes from $\approx 14,000$ to ≈ 443 . Without this reduction the multi-scenario sweeps and candidate-evaluation procedures would be out of reach (§3.2).

1.4 Outline

§2 surveys related literature. §3 lays out the model. §4 describes data and calibration. §5 presents results. §6 conducts sensitivity and robustness analyses. §7 discusses policy implications and limitations. §8 concludes.

2 Literature Review

2.1 Forest-residue and SAF supply-chain optimization

The engineering-OR literature on biomass-to-jet supply chains is well developed (You et al., 2012; Sharma et al., 2013; Lin et al., 2014; Marufuzzaman et al., 2014; Roni et al., 2017). These models share a fixed selling price for the biofuel output, deterministic yields, and cost minimization as the sole objective. Risk and welfare, where addressed, enter as ex-post sensitivity rather than as part of the decision.

2.2 Demand elasticity for jet fuel and SAF

Two distinct elasticities show up in this literature. *Passenger air travel* demand is moderately elastic: Brons et al. (2002) report a meta-analytic mean own-price elasticity of -1.146 across 37 studies, and the InterVISTAS/IATA (2007) benchmarks are -0.6 (pan-national), -0.8 (national), and -1.4 (route). *Jet fuel* as a derived input is much less elastic: Ryu and Janzen (2024) document a 9.5%/0.6% price-quantity ratio implying a descriptive elasticity near -0.063 .

Direct empirical estimates for SAF are scarce; the voluntary market is too young and too thin. The closest analog is Brazilian fuel ethanol — a mandate-driven biofuel with a fossil substitute. Cardoso et al. (2019), with an IV-based estimation under the flex-fuel-vehicle regime, report own-price elasticities of ≈ -1.5 for ethanol and -0.9 for gasoline; Salvo and Huse (2013) provide complementary revealed-preference evidence on gasoline-ethanol substitution. We adopt a baseline $\varepsilon_{\text{SAF}} = -0.5$ with a sensitivity grid $\{-0.3, -0.5, -1.0\}$, defended in §4.4.

2.3 Stochastic programming for biofuel investment

Two-stage stochastic programming has been applied to biorefinery siting (Huang et al., 2010; Tong et al., 2014; Zhang and Hu, 2013), typically with small scenario sets (≤ 10) and standard Benders or L-shaped decomposition. Our scale (20 KDE scenarios, $\sim 9,660$ first-stage binaries,

$\sim 700,000$ second-stage continuous variables per scenario) sits at the edge of both: the extensive-form deterministic equivalent fails to converge (§5.5), and integer Benders would carry a master MILP with the full binary set plus 20 accumulating cuts per iteration. We use a candidate-evaluation procedure (§5.5) instead, which shares Benders’ LP-tight subproblem oracle but evaluates it on a hand-picked candidate set rather than iterating a master.

2.4 Welfare framework

The Marshallian decomposition of total surplus into consumer surplus (CS), producer surplus (PS), and externality terms is the workhorse framework of welfare economics (Pigou, 1920; Just et al., 2004). Its application to renewable fuel markets traces to Ando and Khanna (2000), de Gorter and Just (2009), and Khanna et al. (2017).

A parallel literature shows that biofuel policies can have ambiguous welfare and emissions effects when feedstock production induces indirect land-use change (Searchinger et al., 2008; Hertel et al., 2010) or when the policy instrument creates intra-industry rent transfers (Holland et al., 2009; Lade and Bushnell, 2019). Our scope sets these channels aside. Forest harvest residue is a co-product of conventional timber harvest, not new agricultural land, which sidesteps the Searchinger–Hertel ILUC channel. Partial equilibrium forgoes the economy-wide spillovers Hertel et al. emphasize. Deferring carbon-policy instruments (a Pigouvian τ and the 45Z step credit) to a companion paper sidesteps the LCFS rent-transfer mechanisms in Holland et al. and the RFS pass-through structures in Lade and Bushnell. The trade-off is explicit: a comparison with a full economy-wide stochastic CGE is left to follow-up work.

We treat endogenous SAF demand under a social-planner objective:

$$\max W = \int_0^Q P_{\text{SAF}}(q) dq - C(Q), \quad (1)$$

where the integral is the gross consumer benefit and $C(Q)$ is total chain cost. At an interior optimum, $P_{\text{SAF}}(Q^*) = MC(Q^*)$, the standard market-clearing condition.

2.5 Risk-aware optimization

We use the conditional-value-at-risk (CVaR) of Rockafellar and Uryasev (2000), linearized in the usual way with one scalar η and one slack $\zeta_s \geq 0$ per scenario. We apply CVaR to *welfare losses* relative to the wait-and-see optimum rather than to project cost (cf. Conejo et al., 2010 for the cost convention in energy-systems work). For a social planner, welfare loss is the decision-relevant scalar; the policy parameter λ then represents aversion to worst-tail welfare loss directly.

3 Model Formulation

3.1 Two-stage MILP backbone

Let H index forest-residue source nodes (hex centroids, aggregated to counties; see §3.2), D depot candidate sites, and R biorefinery candidate sites. Let A_d and A_r index ash-content categories at depots and refineries, respectively. The first-stage binary variables $\text{DvD}_{d,a} \in \{0, 1\}$ and $\text{DvB}_{r,a'} \in \{0, 1\}$ select depot and refinery sites and ash classes. The second-stage continuous flow variables are

$x_{h,d,a} \geq 0$ (hex-to-depot residue flow within ash class a) and $z_{d,a,r,a'} \geq 0$ (depot-to-refinery flow). Binary placement, capacity, ash-balance, and mass-balance constraints follow Roni et al. (2017) and our prior implementation in Li et al. (2024) and Li et al. (2025); the full set is detailed in Appendix A.

3.2 County-level aggregation methodology

The reference GAMS model uses $\approx 14,040$ hex grid candidates; the resulting extensive-form stochastic program is intractable. We aggregate hexes to U.S. counties (FIPS) using `shapely`-computed area shares, exact under spatial uniformity within each hex. Of the 14,040 hexes, 8,864 (63%) straddle multiple county boundaries, so single-FIPS assignment introduces a bias the area-share method removes. The cost-coefficient renormalization $K_F = c \cdot n_{\text{P-hex/county}} / A_{\text{Avail-Inv/county}}$ keeps the LP objective aggregation-invariant. The aggregated Stage 1 model reproduces the hex-level GAMS reference within $+0.15\%$ on total cost and -0.07% on residue, and Stage 2 binary placement agrees with the hex-level reference at every shared scenario (Pearson $r = 0.998$ on county-level depot probability across the 20-bin price sweep).

3.3 Endogenous SAF demand: a separate market

We treat SAF as economically distinct from fossil Jet-A on the buyer side. The motivation is institutional: ESG-driven corporate offtakes (United 2.9 B gal cumulative, Delta 0.91, American 0.62, SABA ≥ 0.05 per RFP cycle), federal credits (45Z), state low-carbon-fuel programs (California LCFS, Oregon, Washington), and the EU’s ReFuelEU mandate all create demand for *certified* SAF, not for jet fuel as a generic commodity. The framing also avoids a degenerate corner: under a perfect-substitutes objective $W = \int P dq - C$, the social planner picks Jet-A whenever $MC_{\text{SAF}} > P_{\text{fossil}} \approx \$3.50/\text{gal}$, and since $MC_{\text{SAF}} \approx \$5.50/\text{gal}$ in our calibration, the chain is never built. Treating SAF as a separate market lets the price clear at the SAF-specific willingness to pay.

The implementation is light: we replace the fixed-price parameter $PSAF$ and the legacy slack import term with a linear inverse demand $P_{\text{SAF}}(Q) = a - bQ$. Slack — previously a feasibility filler for unmet demand — disappears; supply shortages are absorbed along the demand curve. The objective becomes

$$\min_{x,z,\text{DvD},\text{DvB}} C(\cdot) - I_{\text{CS}}, \quad (2)$$

where $C(\cdot)$ collects discounted harvest, stumpage, transport, capex, and opex flows and $I_{\text{CS}} = \int_0^Q P_{\text{SAF}}(q) dq$ is total consumer benefit, approximated piecewise (§3.4). Maximizing $-(\text{obj})$ recovers $W = \text{CS} + \text{PS}$.

3.4 SOS2 piecewise welfare integral

For linear inverse demand the gross-benefit integral is the quadratic $I_{\text{CS}}(Q) = aQ - bQ^2/2$. We approximate it on $\{q_k, c_k\}_{k=0}^N$ via

$$\sum_{k=0}^N \lambda_k = 1, \quad \lambda_k \geq 0, \quad \text{SOS2}(\lambda_0, \dots, \lambda_N), \quad (3)$$

$$Q = \sum_k q_k \lambda_k, \quad I_{\text{CS}} = \sum_k c_k \lambda_k. \quad (4)$$

The SOS2 enforcement is redundant here (a concave maximand makes the LP relaxation tight) but we keep it for extensibility to non-quadratic demand specifications. The default grid is $N + 1 = 21$ uniform breakpoints on $[0, Q_{\max}]$, where $Q_{\max} = a/b$. Resolution is tested in §6.3.

3.5 Stochastic price uncertainty (KDE)

Price uncertainty enters via a parallel additive shift of the inverse-demand intercept: in scenario s ,

$$a_s = a_{\text{baseline}} + (\text{PSAF}_s - \overline{\text{PSAF}}_{\text{KDE}}), \quad P_{\text{SAF},s}(Q) = a_s - bQ, \quad (5)$$

holding b (and the local elasticity at the shifted baseline) fixed. The bin centers PSAF_s and probabilities p_s come from a 20-bin Gaussian-kernel density estimate of EIA Gulf-Coast jet-fuel wholesale prices, 2014–2024, bandwidth by Silverman’s rule. Table 1 reports the scenario set; the same 20 bins are used in the wait-and-see sweep (§5.2) and the risk-aware analysis (§5.5–5.6).

Table 1: Twenty-bin KDE scenario set for the SAF wholesale price. Bin centers are the within-bin sample means of the underlying EIA wholesale series. The bin-9 center coincides with the median KDE mass and the topological-expansion threshold of §5.2.

Bin s	PSAF range (\$/gal)	PSAF $_s$ (\$/gal)	p_s (%)	$\sum_{s' \leq s} p_{s'}$ (%)
1	3.81 – 4.08	3.99	3.28	3.3
2	4.08 – 4.34	4.22	5.93	9.2
3	4.34 – 4.61	4.47	7.50	16.7
4	4.61 – 4.87	4.73	7.49	24.2
5	4.87 – 5.14	5.00	6.75	30.9
6	5.14 – 5.40	5.28	6.71	37.7
7	5.40 – 5.67	5.53	7.43	45.1
8	5.67 – 5.93	5.82	7.72	52.8
9	5.93 – 6.20	6.05	6.35	59.2
10	6.20 – 6.46	6.34	4.44	63.6
11	6.46 – 6.73	6.61	3.95	67.5
12	6.73 – 7.00	6.88	4.81	72.4
13	7.00 – 7.26	7.11	5.81	78.2
14	7.26 – 7.53	7.40	6.07	84.2
15	7.53 – 7.79	7.66	5.65	89.9
16	7.79 – 8.06	7.92	4.54	94.4
17	8.06 – 8.32	8.17	2.89	97.3
18	8.32 – 8.59	8.45	1.48	98.8
19	8.59 – 8.85	8.70	0.79	99.6
20	8.85 – 9.12	8.94	0.42	100.0
<i>KDE overall mean</i>			\$5.977/gal	

3.6 Risk-aware objective: CVaR over welfare loss

Let $L_s(\text{DvD}, \text{DvB}) = W_s^{\text{WS}} - W_s^{\text{HN}}(\text{DvD}, \text{DvB})$ be the welfare loss relative to wait-and-see, where W_s^{HN} is the second-stage welfare under a non-anticipative first-stage plan re-evaluated in scenario s . A risk-aware planner solves

$$\min_{(\text{DvD}, \text{DvB}) \in \mathcal{Y}} (1 - \lambda) \mathbb{E}_s[L_s] + \lambda \text{CVaR}_\alpha(L_s), \quad \lambda \in [0, 1], \quad (6)$$

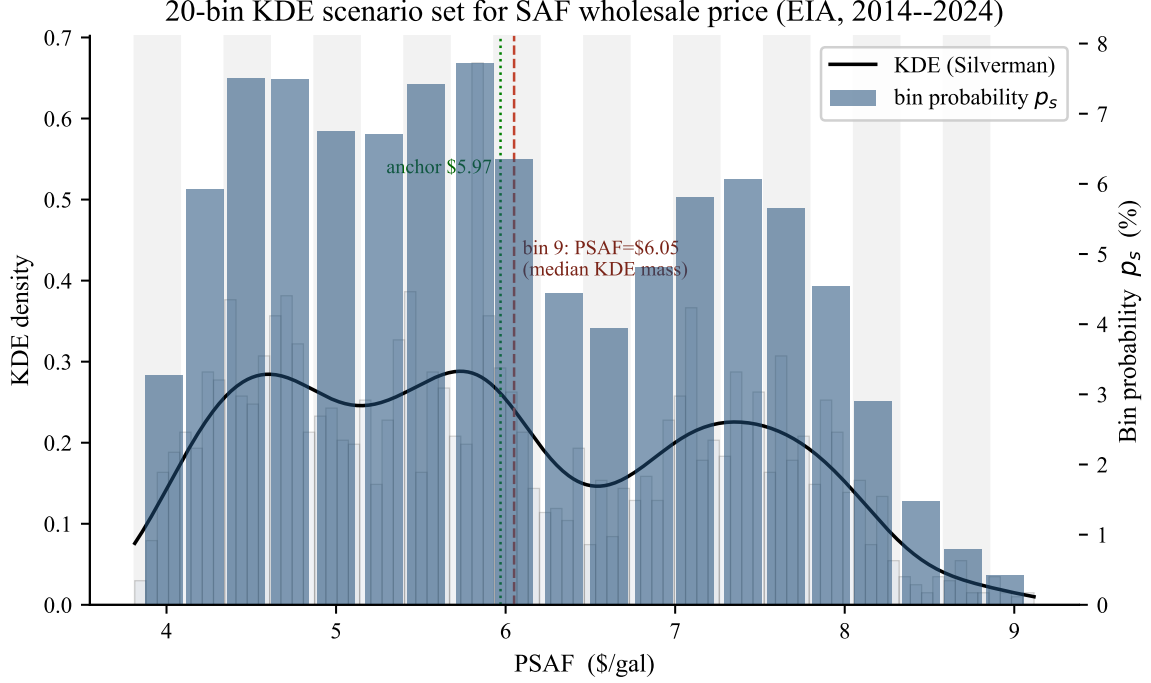


Figure 1: The 20-bin KDE scenario set. Histogram (light grey) shows the underlying daily wholesale-price observations 2014–2024; the smooth black curve is the Silverman-bandwidth Gaussian KDE; blue bars on the secondary axis are the bin probabilities p_s used in the wait-and-see and risk-aware analyses. The bin-9 center coincides with the median KDE mass and with the topological-expansion threshold of \$5.2.

with the Rockafellar–Uryasev representation

$$\text{CVaR}_\alpha(L) = \min_{\eta} \left\{ \eta + \frac{1}{\alpha} \sum_s p_s \zeta_s : \zeta_s \geq L_s - \eta, \zeta_s \geq 0 \right\}. \quad (7)$$

Substituted into (6) this yields a multi-scenario extensive form with $|S|$ copies of the second-stage block, $|S|$ slacks ζ_s , and one shared η . At our scale this is numerically borderline (§5.5). We do not solve the joint MILP. Instead, for any finite candidate set $\mathcal{C} \subseteq \mathcal{Y}$, we fix the binaries to each $y_c \in \mathcal{C}$, evaluate the second-stage LP across all 20 scenarios via the parametric procedure of §5.5, and compute $\mathbb{E}[L_c]$ and $\text{CVaR}_\alpha(L_c)$ exactly. The within- $\mathcal{C}$ minimizer of (6) at any λ is read off the resulting points. Table 5 reports the Pareto-relevant subset for $\mathcal{C} = \{y_1^*, \dots, y_{20}^*\}$ (the 20 wait-and-see candidates); two are undominated. Hybrid plans (e.g. ash-class subsets of the undominated candidates, or non-WS plans) and the joint RU MILP solve are left to follow-up work; the risk premium we report is a within-WS-set statement.

4 Data and Calibration

4.1 Forest residue inventory

Hex-level annual recoverable forest residue is taken from the 2016 Billion-Ton Update (Langholtz et al., 2016) for the southeastern U.S., filtered by harvest type (hardwood saw and pulp residues,

HW only) and aggregated to county. The same hex-level dataset and HW-only filter underlie our prior deterministic analyses (Li et al., 2024; Li et al., 2025), where the data-construction details are reported in full. The aggregate residue used by the chain in our calibration is approximately 25.8 M dry tons/yr, consistent with the more recent 2023 Billion-Ton Report (Langholtz et al., 2024; ORNL/SPR-2024/3103, Chapter 4), which estimates Southeast logging residue plus mill residue plus thinnings near 30 M dt/yr in Year 1 rising to roughly 60 M dt/yr by Year 30 at stumpage prices up to \$70/dt. The hardwood-only filter excludes softwood mill demand (SoftSaw and SoftPulp); relaxing this filter is a natural sensitivity not pursued here. Stumpage costs c_{stump} and harvest costs c_{harv} are hex-level. Transport costs follow Roni et al. (2017), with thresholds $H2D_{\text{max}} = 50$ mi and $D2R_{\text{max}} = 75$ mi.

4.2 Hex-to-county aggregation

The aggregation table `aggregation/hex_county_share.csv` contains for each (hex, FIPS) pair the area share $\phi_{h,F} \in [0, 1]$ with $\sum_F \phi_{h,F} = 1$. Aggregated supply $A_F = \sum_h \phi_{h,F} A_h$ and weighted-average per-unit costs follow naturally; the cost-coefficient renormalization preserves objective value at the aggregated optimum.

4.3 Cost parameters

We use the cost recipe of Roni et al. (2017): depot OPEX = \$0.063/gal-equivalent and depot fixed capital = \$39 M (size-scaled to 200 kdt/yr); per-refinery fixed capital investment = \$1.068 B (size-scaled to a 100 M-gal/yr nth-plant Fischer-Tropsch design), per-refinery variable operating cost = \$1.96/gal; pretreatment recovery RES2PRE = 0.85; discount rate IRR = 10% with a 20-year capital-recovery annuity, reported in single-year-equivalent terms (10-year project NPV, equivalent annual flow). At 34 active refineries and 3.31 B gal/yr the implied total capital investment is roughly \$36 B, or about \$11/annual-gallon, which sits in the upper half of the NREL nth-plant Fischer-Tropsch TEA range \$3.5–\$11/annual-gallon (Tan et al., 2017; Davis et al., 2018; NREL Annual Technology Baseline, 2024). Our biomass-to-SAF yield is calibrated as $\text{BCY} = \text{BCY}_{\text{raw}} / (1 - \overline{\text{ash}}_{\text{baseline}}) = 75.16 / 0.810 = 92.84$ gal/dt on a dry-ash-free basis, with downstream ash-class scale factors RASH $\in [0.83, 0.91]$ delivering effective net yields of 62–67 gal/dt — within the range of nth-plant FT TEAs, and somewhat above demonstration-plant operational data (e.g. LanzaJet ATJ at ~ 36 gal/dt, which is a different pathway). Sensitivity on the slack import cost (now removed from the Path B model but relevant to the §6.2 legacy specification) covers \$2.50–\$5.00/gal. Per-gallon variable operating cost does not separately load hydrogen consumption or Fischer-Tropsch catalyst replacement: in nth-plant TEA convention these are bundled into the variable operating cost row, and our \$1.96/gal sits at the lower end of NREL’s reported range \$2.30–\$2.70/gal (Tan et al., 2017). We treat this as a noted limitation; loading H_2 explicitly would shift the negative-PS gap in §5.3 by approximately \$0.30–\$0.70/gal.

4.4 SAF demand calibration

The baseline equilibrium for Path B is

$$Q^* = 3.0 \times 10^9 \text{ gal/yr}, \quad P^* = \$5.97/\text{gal}, \quad (8)$$

representing the long-run 2030+ scenario in which the SAF Grand Challenge target is met by the supply-chain region. Cross-check: actual 2025 U.S. SAF supply is 0.5–0.7 B gal/yr at a wholesale

price near \$9/gal, a point on the same downward-sloping demand curve (IATA, 2025; EIA, 2025). We adopt baseline $\varepsilon_{\text{SAF}} = -0.5$ with sensitivity $\{-0.3, -0.5, -1.0\}$. Three pieces of evidence support this. SAF is more elastic than baseline jet fuel ($\varepsilon \approx -0.2$) because consumers have a fossil substitute available at \$3.50/gal. The closest empirical analog, Brazilian fuel ethanol, has own-price elasticity ≈ -1.5 (Cardoso et al., 2019). And at the airline pass-through layer, $\varepsilon_{\text{passenger}} \times s_{\text{fuel}} \approx -0.8 \times 0.27 \approx -0.22$ gives a consistency check across the SAF and passenger markets.

The calibration of the inverse demand $P_{\text{SAF}}(Q) = a - bQ$ at (Q^*, P^*) given ε uses

$$b = -\frac{P^*}{\varepsilon Q^*}, \quad a = P^* + bQ^*, \quad (9)$$

which gives $(a, b, Q_{\text{max}}) = (\$25.87/\text{gal}, 6.63 \times 10^{-9}, 3.9 \times 10^9)$ for $\varepsilon = -0.3$, $(\$17.91, 3.98 \times 10^{-9}, 4.5 \times 10^9)$ for $\varepsilon = -0.5$, and $(\$11.94, 1.99 \times 10^{-9}, 6.0 \times 10^9)$ for $\varepsilon = -1.0$. The 21 SOS2 breakpoints per elasticity are tabulated in `python_model/stage2_out/path_b_calibration.csv`.

4.5 KDE price distribution

The 20-bin Gaussian KDE is fit to EIA Gulf-Coast jet-fuel wholesale prices over 2014–2024 with Silverman’s rule. The bin centers span \$3.99–\$8.94/gal and the KDE median falls near \$5.977/gal, which is taken as the central-scenario reference; the demand anchor $P^* = \$5.97$ in §4.4 was chosen independently from techno-economic considerations and rounds to two decimals at this point.

5 Results

5.1 Deterministic baseline (fixed PSAF; Path A reference)

With PSAF held fixed at \$5.97/gal and demand at `TotalBiofuel` = 3×10^9 gal/yr (the pre-Path B specification), the deterministic optimum is summarized in Table 2.

Table 2: Deterministic baseline (Path A reference, PSAF = \$5.97/gal).

Metric	Value
Project value cost (PVC)	\$1.146 B
Depots placed	233
Biorefineries placed	32
Total residue used	25.8 M dt/yr
Biofuel produced	2.97 B gal
Slack imported	0.03 B gal
Solver gap	23.88% (1800 s, 5% target)

This serves as the reference topology against which Path B endogenous-demand and Path A risk-aware results are compared.

5.2 Wait-and-see scenario sweep

Under the 20-bin KDE price sweep, each scenario is solved as an independent deterministic problem with PSAF fixed at the bin center, terminated at a 5% optimality-gap target. The KDE-weighted expected project cost over the resulting incumbents is

$$\mathbb{E}[\widehat{\text{PVC}_{\text{ws}}}] = \$1.541 \text{ B/yr}, \quad (10)$$

which we treat as an upward-biased estimate of the true wait-and-see expected cost (we use this caveat in §5.5 when discussing the recourse-problem bound). The chain-versus-import break-even price is approximately \$5.83/gal: below this, the social cost of building the chain exceeds the cost of meeting demand via imported fossil Jet-A. At PSAF = \$6.05/gal we observe a topological expansion threshold — the optimal placement count jumps from 227 to 248 depots, corresponding to a discrete activation of the second tier of marginal sites. This threshold coincides with the median KDE mass, providing a geometrically natural reference point that we use in §5.6 to motivate the central first-stage candidate for the welfare-loss risk analysis.

5.3 Endogenous-demand equilibrium (Path B)

We solve the Path B MILP with 21 SOS2 breakpoints and a MIP-start initialized from the wait-and-see solution at PSAF = \$6.05/gal (the median KDE bin and topological-expansion threshold of §5.2). The warm-start strategy is essential: a cold-start run at $\varepsilon = -0.5$ with 11 breakpoints and 600 s time limit terminates with a 283% optimality gap, while the warm-started 21-breakpoint run converges to $\leq 5\%$ in under 30 seconds across the elasticity grid. Table 3 reports the social-planner welfare equilibrium at each elasticity, with all dollar figures in single-year discounted equivalents (IRR = 10%, $\delta = 1/(1 + \text{IRR})$).

Table 3: Path B endogenous-demand welfare equilibrium across the elasticity grid. All dollar figures are single-year discounted at IRR = 10%. Q^* is essentially invariant in ε because the residue supply ceiling binds.

ε	Q^* (B gal)	P^* (\$/gal)	CS (\$B)	PS (\$B)	W (\$B)	Dep.	Ref.
−0.3	3.304	3.95	32.92	−4.89	28.03	248	34
−0.5	3.314	4.72	19.86	−2.57	17.27	248	34
−1.0	3.315	5.34	9.94	−0.70	9.23	248	34

Three observations are worth highlighting. First, Q^* is essentially invariant across the elasticity grid (3.304–3.315 B gal/yr) and the same first-stage topology of 248 depots and 34 biorefineries is selected at every elasticity. This indicates the model operates in a *supply-constrained regime* at the baseline calibration: the marginal welfare gain of producing an additional gallon exceeds its marginal cost across the entire elasticity range, and what binds the optimum is the residue supply ceiling — not the marginal-revenue-equals-marginal-cost condition. The interpretation is that, conditional on the SAF Grand Challenge demand environment, regional residue availability rather than buyer-side price sensitivity is the binding economic constraint on chain scale.

Second, P^* moves systematically with ε (\$3.95 $\rightarrow$ \$4.72 $\rightarrow$ \$5.34 as $|\varepsilon|$ rises from 0.3 to 1.0). Mechanically this is the inverse demand curve evaluated at the (roughly fixed) Q^* : more elastic demand is geometrically flatter, so it lifts more under a fixed-quantity slice. Economically, the social planner sets P^* at the height of the demand curve at Q^* , capturing the full marginal-buyer surplus.

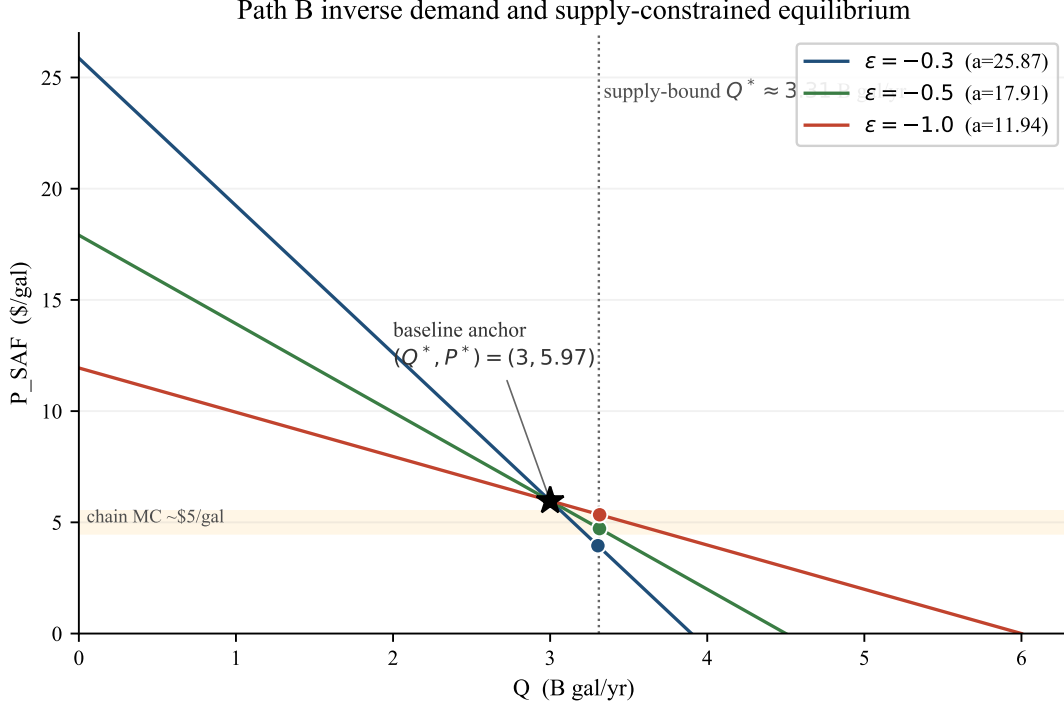


Figure 2: Path B inverse demand and supply-constrained equilibrium. The three colored lines are $P_{\text{SAF}}(Q) = a - bQ$ at the calibration elasticities $\varepsilon \in \{-0.3, -0.5, -1.0\}$, all anchored at $(Q^*, P^*) = (3.0 \text{ B}, \$5.97/\text{gal})$ (black star). The vertical dotted line marks the supply-bound $Q^* \approx 3.31 \text{ B gal/yr}$ selected by the social planner under all three calibrations; the colored circles are the equilibrium (Q^*, P^*) points where the demand curves cross this supply ceiling. The orange band indicates the chain marginal-cost range; equilibrium prices fall *below* this band, generating the negative producer surplus reported in Table 3.

Third, producer surplus is *negative* at every ε , ranging from $-\$0.70 \text{ B/yr}$ ($\varepsilon = -1.0$) to $-\$4.89 \text{ B/yr}$ ($\varepsilon = -0.3$). This is the welfare-economics signature of a social-planner solution: by pushing P^* down to the demand curve at Q^* (rather than to the chain’s average cost), the planner trades producer revenue for consumer surplus until the marginal-welfare condition holds. The implied per-gallon revenue gap is $\$0.21/\text{gal}$ at $\varepsilon = -1.0$ and $\$1.48/\text{gal}$ at $\varepsilon = -0.3$, scaling roughly with $1/|\varepsilon|$. We interpret this as a quantitative reference for the IRA Section 45Z producer credit, which, under prevailing-wage compliance, pays

$$\text{Credit}_{45Z} = \$1.75/\text{gal} \times \max\left(0, \frac{50 - \text{CI}_{\text{fuel}}}{50}\right), \quad (11)$$

with CI_{fuel} in kg $\text{CO}_2\text{e/MMBtu}$ and a baseline of 50 kg $\text{CO}_2\text{e/MMBtu}$ (IRA §13704; Treasury 1.45Z-1 NPRM, January 2025). For a Fischer-Tropsch forest-residue pathway with $\text{CI} \approx 5\text{--}11 \text{ kg CO}_2\text{e/MMBtu}$ (CORSIA defaults), Eq. (11) delivers a credit of roughly $\$1.38\text{--}\$1.56/\text{gal}$ — which falls within our $\varepsilon = -0.3$ planner-PS-gap range and overshoots the $\varepsilon = -1.0$ range. We caution that the alignment is qualitative rather than identifying: the planner’s per-gallon gap reflects $\text{AC}(Q^*) - P_{\text{planner}}^*$ where P_{planner}^* is the demand-curve height at Q^* , while 45Z compensates a private producer at the competitive-equilibrium price $P_{\text{competitive}}^*$ that may differ. A formal derivation of the planner-to-private mapping is an explicit task for the companion Paper 2.

5.4 Welfare decomposition

Following Just, Hueth, and Schmitz (2004) we decompose total surplus into consumer and producer components:

$$CS^* = \frac{1}{2}(a - P^*)Q^*, \quad PS^* = P^*Q^* - C(Q^*), \quad (12)$$

with values reported in Table 3 and visualized in Figure 3. The CS-to-|PS| ratio rises sharply with the inverse-demand intercept (steepness): from 14.2 at $\varepsilon = -1.0$ to 6.7 at $\varepsilon = -0.3$, indicating that — to our knowledge for the first time in the SAF literature — the welfare distribution between buyers and producers depends almost entirely on the demand-side elasticity, while the chain topology and produced quantity stay nearly fixed (§5.3). The supply-constrained regime is therefore an unusually clean setting for welfare attribution: any policy intervention that does not relax the supply ceiling primarily redistributes between CS and PS, leaving total W second-order in topology choice. We pick this point up in the policy discussion of §7.1.

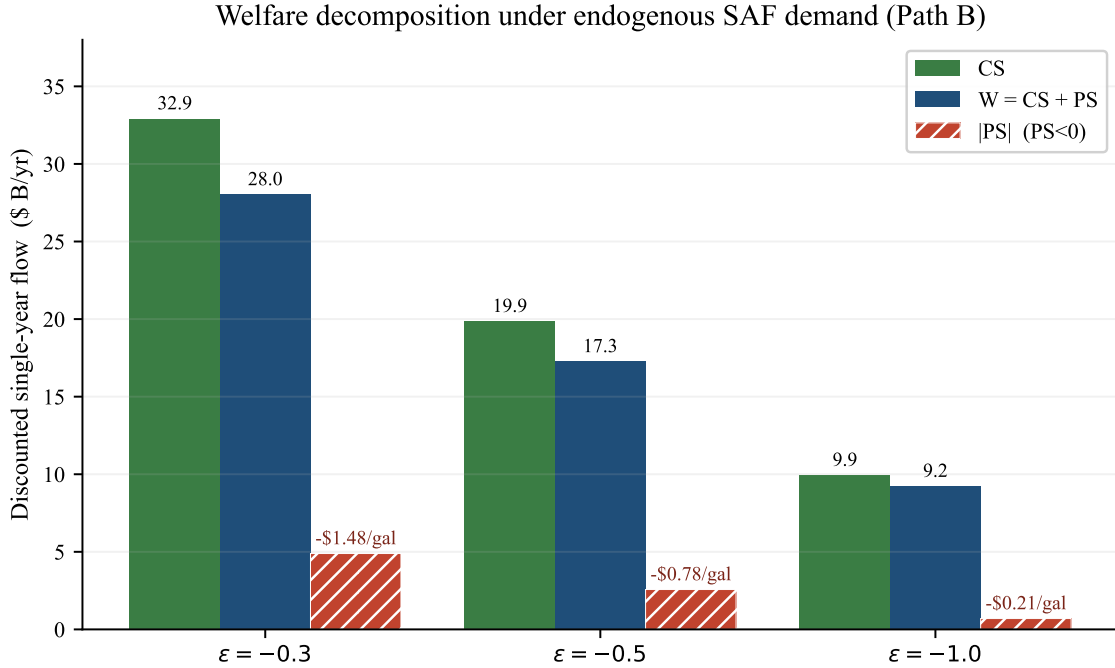


Figure 3: Welfare decomposition under Path B endogenous demand. Green bars show consumer surplus, blue bars total welfare $W = CS + PS$, hatched red bars the magnitude of producer surplus $|PS|$ (PS is negative at every elasticity). The per-gallon revenue-deficit annotations on the $|PS|$ bars equal $|PS|/Q^*$ and serve as a qualitative reference for the IRA Section 45Z producer credit (Eq. 11); the precise mapping between the planner’s per-gallon gap and the 45Z credit a private producer would require is treated formally in the companion Paper 2.

5.5 Quick-Win upper bound on the recourse-problem expected cost

Two attempts at the extensive-form deterministic equivalent of the recourse problem failed to converge. The barrier algorithm stalled at numerical noise floor 2×10^{-3} residual; branch-and-bound never started. We therefore propose the following Quick-Win procedure, which we use to bound the recourse-problem expected cost from above without solving the joint problem.

Let y_c^* denote the wait-and-see first-stage solution under scenario $c \in \{1, \dots, 20\}$. Each y_c^* is a *feasible non-anticipative candidate*: a specific first-stage plan that could in principle be committed to without observing the realized scenario. For each candidate we compute its expected cost across the price distribution by fixing binaries to y_c^* and re-evaluating the second-stage LP over all 20 KDE scenarios. Because the binary fix collapses the MILP to a pure LP, we share a single LP build across scenarios and rewrite `z[.Obj]` parametrically per scenario; primal-simplex warm starts each subsequent solve from the previous basis. The full $20 \times 20 = 400$ LP sweep completes in 18.8 minutes wall clock — approximately $50\times$ faster than the failed extensive-form attempts.

By definition, the recourse-problem optimal expected cost satisfies

$$\mathbb{E}[\text{PVC}_{\text{RP}^*}] = \min_y \mathbb{E}_s[\text{PVC}(y, s)] \leq \min_{c \in \{1, \dots, 20\}} \mathbb{E}[\text{PVC} \mid y_c^*], \quad (13)$$

since the wait-and-see candidate set is a (finite) subset of the feasible non-anticipative plans. Computing the right-hand side from the LP sweep yields

$$\mathbb{E}[\text{PVC}_{\text{RP}^*}] \leq \min_c \mathbb{E}[\text{PVC} \mid y_c^*] = \$0.925 \text{ B/yr}, \quad (14)$$

achieved at $c^* = 20$ (the highest-PSAF wait-and-see plan, 278 depots and 44 refineries). This is an upper bound on $\mathbb{E}[\text{PVC}_{\text{RP}^*}]$ established without an extensive-form or Benders solve.

We do not claim a corresponding tight lower bound. The obvious candidate is $\mathbb{E}[\text{PVC}_{\text{WS}}] = \1.541 B/yr , valid as a lower bound in theory; in our data it was computed from 20 deterministic MIPs at a 5% gap target and carries upward bias of at least $\$616 \text{ M/yr}$ (the violation $\mathbb{E}[\text{PVC}_{\text{WS}}]_{\text{reported}} > \min_c \mathbb{E}[\text{PVC} \mid y_c^*]$ would otherwise be impossible). Tightening this would require re-solving the 20 sweep MIPs to negligible gap (estimated 100–200 CPU-h), which we do not undertake. We therefore report $\$0.925 \text{ B/yr}$ as a numerically conservative upper bound on $\mathbb{E}[\text{PVC}_{\text{RP}^*}]$ and treat EVPI as an open quantity.

On Benders. We did not implement integer Benders or L-shaped decomposition. The master MILP retains all 9,660 first-stage binaries plus 20 optimality cuts per iteration; the deterministic Path B MILP at the same binary scale converges to a 5%-gap solve in roughly 30 seconds with a wait-and-see warm start, but per-iteration Benders masters slow down systematically as cuts accumulate, requiring a few minutes per master by mid-iteration in pilot sketches. With 5–15 iterations typical at this size and ~ 5 minutes per iteration of subproblem work, the wall budget is on the order of hours to a day — feasible but expensive relative to the 18.8-minute candidate sweep, and not the priority here. The candidate-evaluation upper bound shares Benders’ LP-tight subproblem oracle but skips the master iteration; a converged Benders run would also have to handle the same gap-bound wait-and-see lower bound, so the EVPI tightening still requires re-solving the sweep at low gap.

Two side observations are independent of the EVPI question. First, $\mathbb{E}[\text{PVC} \mid y_c^*]$ is not monotone in c : the smallest expected cost is at $c = 20$ and the second-smallest at $c = 18$, consistent with “larger topology is more robust to upside scenarios.” Second, the topological-expansion threshold of $\$5.2$ (the jump from 227 to 248 depots between bins 8 and 9 at $\text{PSAF} \approx \$6/\text{gal}$) coincides with the median KDE mass; $\$5.6$ uses the bin-9 wait-and-see plan as a natural reference point on these grounds, not as an optimum.

5.6 Risk-aware welfare-loss analysis

We embed the welfare model in the price-uncertainty environment of §5.5 via the parallel-shifted inverse demand $a_s = a_{\text{baseline}} + (\text{PSAF}_s - \overline{\text{PSAF}}_{\text{KDE}})$ with slope b fixed. For each scenario s we solve the Path B MILP twice: once as wait-and-see (W_s^{WS}), and once with binaries fixed to the central candidate $y_{\text{bin } 9}^*$ (248 depots, 34 refineries; the topological-expansion-threshold plan of §5.2) to yield W_s^{HN} . The loss is $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$. Solver time was 37 minutes wall over all 20 scenarios; WS gaps were $\leq 5\%$ in 18 of 20 scenarios and $\leq 8\%$ in all 20.

Three things summarize Table 4. The expected welfare loss from holding the central plan fixed is small in absolute terms: $\mathbb{E}[L] = \$98 \text{ M/yr}$, 0.56% of expected wait-and-see welfare. The loss is concentrated: $L_s = 0$ exactly in 17 of 20 bins (cumulative probability 81.7%), and the entire $\mathbb{E}[L]$ comes from three non-contiguous bins $\{1, 3, 4\}$. The tail is bigger than the mean: $\text{CVaR}_{0.05}(L) = \723 M/yr , $\approx 7.4 \times \mathbb{E}[L]$, driven by bin 1 ($L_1 = \776 M/yr, $p_1 = 0.033$) where the WS planner would have downsized to 193 depots and 27 refineries; bins 3 and 4 contribute \$623 and \$341 M/yr at probability 0.075 each.

Table 4: Path B risk-aware welfare-loss analysis under price uncertainty. Wait-and-see (W_s^{WS}) and here-and-now (W_s^{HN}) welfares are computed under parallel-shifted inverse demand $a_s = a_{\text{base}} + (\text{PSAF}_s - \overline{\text{PSAF}}_{\text{KDE}})$. $\text{CVaR}_{0.05}$ averages over the worst-5% probability tail. Dollar figures in single-year discounted equivalents at $\text{IRR} = 10\%$.

Quantity	Value
$\mathbb{E}[W^{\text{WS}}]$	\$17.48 B/yr
$\mathbb{E}[W^{\text{HN}} \mid y_{\text{bin } 9}^*]$	\$17.38 B/yr
$\mathbb{E}[L]$	\$ 0.098 B/yr
$\text{CVaR}_{0.05}(L)$	\$ 0.723 B/yr
$\text{CVaR}_{0.05}(W^{\text{HN}})$	\$11.72 B/yr
L_s range across scenarios	[\$0, \$0.776 B/yr]
W_s^{WS} range	[\$12.15, \$26.21 B/yr]

A full 20-candidate Pareto evaluation sharpens the picture. Treating each y_c^* as a non-anticipative first-stage candidate, we run the HN sweep over all 20 scenarios for every c via the parametric procedure of §5.5, computing $\mathbb{E}[L_c]$ and $\text{CVaR}_{0.05}(L_c)$ at each. The 17 additional candidates take 48 minutes wall on a single workstation; the Pareto-relevant subset is in Table 5.

Table 5: Pareto-relevant first-stage plans from the full 20-candidate evaluation. The first two rows are Pareto-undominated; the third is the central plan from §5.3, dominated by bin 6 on both axes; the last two are wait-and-see corners.

Plan (PSAF anchor)	Topology dep / ref	$\mathbb{E}[L]$ \$ M/yr	$\text{CVaR}_{0.05}(L)$ \$ M/yr	Note
$y_{\text{bin } 6}^*$ (\$5.28)	225 / 31	92.7	435.6	risk-neutral best
$y_{\text{bin } 8}^*$ (\$5.82)	227 / 31	99.8	427.8	$\text{CVaR}_{0.05}$ best
$y_{\text{bin } 9}^*$ (\$6.05)	248 / 34	97.7	723.4	central; dominated
$y_{\text{bin } 1}^*$ (\$3.99)	193 / 27	472.2	1,909.5	smallest WS
$y_{\text{bin } 20}^*$ (\$8.94)	278 / 44	1,161.5	2,529.0	largest WS

Two candidates are Pareto-undominated. A planner minimizing $(1 - \lambda) \mathbb{E}[L] + \lambda \text{CVaR}_{0.05}(L)$ picks

$y_{\text{bin } 6}^*$ for $\lambda < 0.48$ and $y_{\text{bin } 8}^*$ for $\lambda > 0.48$. The risk premium — the extra expected loss a fully CVaR-averse planner accepts relative to a risk-neutral one — is $\Delta\mathbb{E}[L] = \$7.1 \text{ M/yr}$, in exchange for $\Delta\text{CVaR}_{0.05}(L) = -\7.8 M/yr . Both are well below 0.1% of $\mathbb{E}[W^{\text{WS}}]$ in absolute terms but the trade-off is non-degenerate. The CVaR-optimal plan is one depot larger (227 vs. 225, same 31 refineries) and matches the wait-and-see optimum at the break-even price $\$5.83/\text{gal}$; the risk-neutral plan matches the WS optimum at a slightly lower PSAF.

The dominance of $y_{\text{bin } 6}^*$ over the central plan $y_{\text{bin } 9}^*$ of $\$5.3$ is consequential. The deterministic equilibrium under the median-PSAF anchor selects 248 depots and 34 refineries, but under the realized KDE distribution this is not the welfare-loss minimum: the 23 extra depots and 3 extra refineries pay off only in supply-constrained high-PSAF scenarios (cumulative probability $\approx 25\%$), while their amortized capex is sunk in every scenario. The lower-PSAF mass dominates, so the smaller bin-6 plan improves both $\mathbb{E}[L]$ and $\text{CVaR}_{0.05}(L)$. The deterministic plan over-states the worst-tail loss by $1.66\times$ relative to bin 8; the corresponding $\$300 \text{ M/yr}$ is the cost of using the deterministic plan in a stochastic decision problem.

One caveat. The bin-2 wait-and-see solve terminated at 7.31% gap (vs. $\leq 5\%$ elsewhere), which biases W_2^{WS} low and consequently inflates L_2 at every fixed-binary candidate. The conservative correction $\hat{W}_s^{\text{WS}} = \max(W_s^{\text{WS}}, \max_c W_s^{\text{HN},c})$ raises $\mathbb{E}[L]_{\text{bin } 6}$ by $\approx \$5 \text{ M/yr}$ (within the dominance margin) and leaves the Pareto-undominated set unchanged. Total wall time for the full risk-aware analysis was 85 minutes.

6 Sensitivity and Robustness

6.1 Demand elasticity

The full sweep $\varepsilon \in \{-0.3, -0.5, -1.0\}$ is in Table 3. In the supply-constrained regime, topology and Q^* are essentially insensitive to ε while P^* tracks the demand curve at the supply-bound Q^* , and the producer-surplus deficit scales as $1/|\varepsilon|$. We adopt $\varepsilon = -0.5$ as headline baseline (§4.4); the other two values bound the credible range.

6.2 Slack import cost (legacy specification)

Path B removes slack from the model (SAF and Jet-A are buyer-segmented; §3.3), but slack-import cost remains relevant for the legacy Path A interpretation in which the buyer can fall back to fossil at $\$3.50/\text{gal}$. A sweep over $[\$2.50, \$5.00]/\text{gal}$ would require five additional 20-bin KDE re-runs (≈ 75 CPU-hours) and is not undertaken here. The qualitative direction is straightforward: a higher slack cost raises the chain-versus-import break-even price (more SAF is built at a given PSAF). The shift is roughly one-for-one at high-PSAF scenarios and less than one-for-one at low-PSAF scenarios because of capex thresholds.

6.3 SOS2 breakpoint resolution

A cold-start run with $N + 1 = 11$ breakpoints terminates at a 283% gap in 600s with Q^* snapped to $k = 1$, a clear discretization artifact. With 21 breakpoints and a wait-and-see warm start the run converges to $\leq 5\%$ gap in under 30s. Table 6 tests robustness in two regimes. At the supply-constrained baseline of $\$5.3$ the 21- and 31-breakpoint runs are numerically identical, but

this is degenerate: Q^* is pinned at the residue-supply ceiling and is independent of the SOS2 grid resolution at the optimum. The substantive test is the demand-constrained regime, where Q^* sits in the interior of the grid; we use the bin-1 demand shift as a representative case. The 21- and 31-breakpoint runs agree on welfare within 0.86% (\$12.193 vs. \$12.298 B/yr) and topology within ± 1 depot/refinery; the gap closes faster at 31. We adopt 21 as default; 31 is a tighter approximation in the interior regime.

Table 6: SOS2 breakpoint robustness across the two regimes. The supply-constrained regime is degenerate (grid resolution is irrelevant when Q^* is at a chain-side constraint boundary). The demand-constrained regime is the substantive test; 21 breakpoints recover 31-breakpoint results within $\sim 1\%$ on welfare and ± 1 on facility counts.

Regime	$N+1$	W (\$B)	Q^* (B gal)	dep/ref	Gap
Supply-constr. ($\varepsilon = -0.5$ baseline)	21	17.272	3.314	248 / 34	3.06%
	31	17.272	3.314	248 / 34	3.06%
Demand-constr. (bin-1 shift)	21	12.193	2.446	182 / 26	3.70%
	31	12.298	2.450	183 / 25	1.41%

6.4 KDE bandwidth

The 20-bin KDE uses Silverman’s rule. To check bandwidth robustness we re-fit at $\{0.5\times, 1.0\times, 1.5\times\}$ Silverman, re-integrate over the same bin edges, and re-weight three downstream quantities (Table 7). No MILP re-solves are needed: the binary fix decouples the second-stage LP.

Table 7: Bandwidth sensitivity of the headline stochastic deliverables. The $1.0\times$ row is the canonical specification used elsewhere. The optimum-bound candidate stays at bin 20 under all three bandwidths.

Bandwidth mult. Silverman	$\mathbb{E}[\widehat{\text{PVC}}_{\text{WS}}]$ (\$ B/yr)	QW UB on $\mathbb{E}[\text{PVC}_{\text{RP}^*}]$ (\$ B/yr)	Best cand.	$\mathbb{E}[L]$ (\$ M/yr)	$\text{CVaR}_{0.05}(L)$ (\$ M/yr)
$0.5\times$	1.491	0.867	bin 20	100.2	716.0
$1.0\times$	1.549	0.934	bin 20	97.3	724.4
$1.5\times$	1.631	1.026	bin 20	93.8	731.7

The Path B welfare-loss quantities are more bandwidth-robust than the legacy Path A wait-and-see expectation. The $\text{CVaR}_{0.05}$ tail loss varies by less than 1% across the grid: bin 1 carries similar probability mass under all three smoothing choices, so the CVaR aggregator picks up about the same dollar contribution regardless. The Quick-Win bound varies by $\pm 8\%$ around \$0.934 B/yr, with the optimum candidate stable at bin 20. The legacy $\mathbb{E}[\text{PVC}_{\text{WS}}]$ is the most bandwidth-sensitive but, as in §5.5, also the most upward-biased and not a headline result.

7 Discussion

7.1 Policy implications

Four points stand out.

The Path B equilibrium sits in a *supply-constrained regime* at the SAF Grand Challenge calibration: residue availability, not buyer-side elasticity, sets the produced quantity, and the same 248-depot/34-refinery topology emerges at every elasticity in our grid. Policy levers that do not relax the supply ceiling — price floors, voluntary-market subsidies, blender mandates — mainly redistribute between consumer and producer surplus. Levers that *do* relax it — targeted feedstock procurement, residue-recovery efficiency, depot-network densification on the supply tail — have first-order welfare leverage.

Producer surplus is robustly negative (Table 3). The implied \$0.21–\$1.48/gal revenue gap is comparable in magnitude to the IRA 45Z credit $\$1.75 \times (50 - \text{CI})/50$ (Eq. 11): for low-CI Fischer–Tropsch residue ($\text{CI} \approx 5\text{--}11$ kg CO₂e/MMBtu) the credit is \$1.38–\$1.56/gal, near the upper end of our planner-gap range. The alignment is qualitative, not identifying: the planner’s gap reflects average-cost pricing at the welfare-maximizing Q^* , while 45Z compensates competitive private producers (§5.3).

The Quick-Win upper bound of \$0.925 B/yr on the recourse expected cost shows that a planner who commits without scenario information can achieve expected project cost in the same order of magnitude as wait-and-see. The optimum-bound candidate is the large-topology plan (278 depots, 44 refineries); smaller WS-optimal plans at low PSAF carry meaningfully higher expected cost.

The topological-expansion threshold at PSAF $\approx$ \$6.05/gal sits at the median KDE mass, giving a natural reference point for first-stage candidate selection (§5.6). Targeted policy at that tier — depot subsidies near the supply-ceiling boundary — has high welfare leverage both deterministically and stochastically.

7.2 Limitations

Three limitations are worth flagging. SAF as a separate market is defended on policy grounds (§3.3) but is not currently supported by a direct empirical SAF demand estimate; we return to this in a companion study as voluntary-market data accumulate. Linear inverse demand is a local approximation; a CES or log-linear specification would change the Q^* , P^* numbers but not the welfare decomposition. The present model has no carbon externality: the Pigouvian τ counterfactual and 45Z step-function analysis are the natural extensions but belong in Paper 2.

7.3 Comparison with prior literature

Engineering-OR biomass-to-jet. The closest antecedents on the model-structure axis are You et al. (2012), Sharma et al. (2013), Lin et al. (2014), Marufuzzaman et al. (2014), and Roni et al. (2017), the last of these being the model from which our county-aggregated formulation descends. They share an exogenous SAF price, a demand floor, and a cost-minimization objective; none can decompose project value into consumer surplus, producer surplus, or externality components, since demand is not price-responsive. Our Path B replaces the demand floor with an inverse-demand curve and the fixed price with $P^* = a - bQ^*$, enabling the welfare decomposition of §5.4 and the negative-PS finding of §5.3. This is, to our knowledge, the first welfare-decomposed analysis of the residue-to-jet chain at regional MILP scale.

Stochastic programming for biofuel investment. The closest antecedents on the uncertainty axis are Huang et al. (2010), Zhang and Hu (2013), and Tong et al. (2014), with scenario sets of order 5–10 and standard L-shaped or Benders decomposition. At our scale (20 KDE scenarios, $\sim 9,660$

binaries) the extensive form stalls at numerical noise (§5.5), and integer Benders is feasible but expensive (master retains all binaries plus accumulating cuts; §5.5). The candidate-evaluation upper bound we propose uses the same LP-tight subproblem oracle that Benders would, but evaluates it on a hand-picked set rather than through master iteration. We are not aware of prior biofuel-SP papers reporting a constructive recourse-problem upper bound at this scale.

Emerging SAF-demand literature. The closest contemporary work is the MIT thesis of Membreno (2024), which estimates passenger air-travel elasticities from U.S. DB1B and BTS data and uses them to calibrate scenarios for SAF scale-up. Three differences. Membreno’s elasticities are upstream passenger demand; Path B requires the derived SAF demand, which we calibrate via pass-through and a Brazilian-ethanol analog (§4.4). Membreno’s exercise treats SAF as quantity scale-up at exogenous price; we treat it as a buyer-segmented market with its own demand curve — a stronger commitment that lets us report welfare decompositions and policy-credit thresholds. Our risk-aware analysis (§5.6) and supply-constrained-regime finding (§5.3) hinge on the chain-side discrete capex ladder, which is absent from the upstream-passenger framing. The two are complementary: a SAF-specific empirical demand estimate plugged into the welfare and risk machinery developed here would address the principal limitation of §7.2.

8 Conclusion

We presented a county-aggregated two-stage stochastic MILP for a southeastern U.S. forest-residue-to-SAF supply chain, embedded in a partial-equilibrium welfare framework with endogenous SAF demand, and a candidate-evaluation upper bound on the recourse expected cost that sidesteps the extensive-form and Benders solves. The chain-versus-import break-even SAF price is \$5.83/gal; the topological-expansion threshold sits at \$6.05/gal, the median KDE mass. The candidate-evaluation upper bound is \$0.925 B/yr, attained by the highest-PSAF WS plan (278 depots, 44 refineries). The welfare equilibrium is $W = \$17.27 \text{ B/yr}$ at $\varepsilon_{\text{SAF}} = -0.5$, with a robustly negative producer surplus implying a per-gallon production-credit gap of \$0.78/gal — comparable in magnitude to the IRA Section 45Z credit. The 20-candidate Pareto frontier has two undominated plans, $y_{\text{bin } 6}^*$ and $y_{\text{bin } 8}^*$, with an implied risk premium of \$7.1 M/yr; the deterministic Path B equilibrium plan is dominated on both axes, and the deterministic-vs-stochastic gap on the worst-5% tail is \$300 M/yr. The principal limitation — that demand calibration relies on a pass-through argument rather than a direct SAF-specific empirical estimate — is the natural target for follow-up as voluntary-market data accumulate.

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A Full Path B MILP formulation

This appendix lists the complete decision problem as implemented in the Path B Gurobi source (see Appendix C for the code listing). Constraints (S1)–(S12) match the chain backbone of Roni et al. (2017) modulo the county-aggregation re-normalization of §3.2. The Path B endogenous-demand block (S13)–(S16) replaces the legacy demand-floor constraint and the fixed-PSAF revenue term.

A.1 Sets

H	forest-residue source nodes (counties post-aggregation; $ H \approx 443$)
D	depot candidate sites (county FIPS)
R	biorefinery candidate sites (county FIPS)
A_d	ash classes for depots ($ A_d = 10$)
A_r	ash classes for refineries ($ A_r = 10$)
$\mathcal{HD}$	feasible hex–depot pairs satisfying $H2D_{\max} = 50$ mi
$\mathcal{DR}$	feasible depot–refinery pairs satisfying $D2R_{\max} = 75$ mi
K	SOS2 breakpoint indices, $K = \{0, 1, \dots, N\}$, $N + 1 = 21$

A.2 Parameters

AResidue_h	annual recoverable residue at h (dt/yr)
$\text{ARScost}_h, \text{ARHcost}_h$	per-dt stumpage and harvest cost (\$/dt)
$\text{AtcH2D}_{h,d}, \text{AtcD2B}_{d,r}$	per-dt transport cost (\$/dt)
AshStd_h	residue ash content at h (fraction)
StoUnit	global ash multiplier (= 1 in Paper 1)
$\text{DASH}_a, \text{RASH}_{a'}$	ash-class mean and yield-scale factors
BCY	raw conversion yield, gal SAF / dt residue (= 75.16)
RES2PRE	depot pretreatment recovery (= 0.85)
$\text{DCAP}_{\text{LW}}, \text{DCAP}_{\text{UP}}$	depot capacity bounds (dt/yr)
$\text{RCAP}_{\text{LW}}, \text{RCAP}_{\text{UP}}$	refinery capacity bounds (gal SAF/yr)
$\text{CAPEXD}, \text{OPEXD}$	depot capital and operating costs (\$/yr-equiv)
$\text{CAPEXB}, \text{OPEXB}$	biorefinery capital and operating costs (\$/yr-equiv)
$\text{IRR} = 0.10, \delta = 1/(1 + \text{IRR})$	discount factor
$\pi_a^{\text{lw}}, \pi_a^{\text{up}}$	ash-class lower/upper percentile fractions, depot
$\pi_{a'}^{\text{lw}}, \pi_{a'}^{\text{up}}$	ash-class lower/upper percentile fractions, refinery
a, b	inverse-demand intercept and slope (§4.4)
$q_k = k \cdot Q_{\max}/N$	SOS2 breakpoint quantities
$c_k = aq_k - \frac{1}{2}bq_k^2$	SOS2 breakpoint cumulative-CS values

A.3 Decision variables

$x_{h,d,a} \geq 0$	residue flow from h to depot d in ash class a (dt/yr)
$z_{d,a,r,a'} \geq 0$	flow from depot d (ash a) to refinery r (ash a') (dt/yr)
$\text{DvD}_{d,a} \in \{0, 1\}$	depot d active in ash class a
$\text{DvB}_{r,a'} \in \{0, 1\}$	refinery r active in ash class a'
$Q_{\text{SAF}} \geq 0$	total SAF produced and sold (gal/yr)
$\lambda_k \in [0, 1]$	SOS2 weights, $k \in K$
$I_{\text{CS}} \geq 0$	gross consumer benefit $\int_0^Q P_{\text{SAF}}(q) dq$ (\$/yr)

A.4 Constraints

Chain backbone.

(S1) Source-side residue supply.

$$\sum_{(d,a): (h,d) \in \mathcal{HD}, a \in A_d} x_{h,d,a} \leq \text{AResidue}_h, \quad \forall h \in H. \quad (15)$$

(S2)–(S3) Depot capacity.

$$\text{RES2PRE} \cdot \sum_{h: (h,d) \in \mathcal{HD}} x_{h,d,a} \geq \text{DCAP}_{\text{LW}} \cdot \text{DvD}_{d,a}, \quad \forall d \in D, a \in A_d, \quad (16)$$

$$\text{RES2PRE} \cdot \sum_{h: (h,d) \in \mathcal{HD}} x_{h,d,a} \leq \text{DCAP}_{\text{UP}} \cdot \text{DvD}_{d,a}, \quad \forall d \in D, a \in A_d. \quad (17)$$

(S4) Single ash class per depot.

$$\sum_{a \in A_d} \text{DvD}_{d,a} \leq 1, \quad \forall d \in D. \quad (18)$$

(S5)–(S6) Depot ash-class balance.

$$\sum_{h: (h,d) \in \mathcal{HD}} \text{AshStd}_h \cdot \text{StoUnit} \cdot x_{h,d,a} \geq \text{DCAP}_{\text{LW}} \cdot \pi_a^{\text{lw}} \cdot \text{DvD}_{d,a}, \quad (19)$$

$$\sum_{h: (h,d) \in \mathcal{HD}} \text{AshStd}_h \cdot \text{StoUnit} \cdot x_{h,d,a} \leq \text{DCAP}_{\text{UP}} \cdot \pi_a^{\text{up}} \cdot \text{DvD}_{d,a}. \quad (20)$$

($\forall d \in D, a \in A_d$.)

(S7) Depot mass balance.

$$\sum_{(r,a'): (d,r) \in \mathcal{DR}, a' \in A_r} z_{d,a,r,a'} = \text{RES2PRE} \cdot \sum_{h: (h,d) \in \mathcal{HD}} x_{h,d,a}, \quad \forall d \in D, a \in A_d. \quad (21)$$

(S8)–(S9) Refinery capacity.

$$\text{BCY} \cdot \text{RASH}_{a'} \cdot \sum_{(d,a): (d,r) \in \mathcal{DR}} z_{d,a,r,a'} \geq \text{RCAP}_{\text{LW}} \cdot \text{DvB}_{r,a'}, \quad (22)$$

$$\text{BCY} \cdot \text{RASH}_{a'} \cdot \sum_{(d,a): (d,r) \in \mathcal{DR}} z_{d,a,r,a'} \leq \text{RCAP}_{\text{UP}} \cdot \text{DvB}_{r,a'}. \quad (23)$$

(S10) Single ash class per refinery.

$$\sum_{a' \in A_r} \text{DvB}_{r,a'} \leq 1, \quad \forall r \in R. \quad (24)$$

(S11)–(S12) Refinery ash balance.

$$\sum_{(d,a):(d,r) \in \mathcal{DR}} \text{DASH}_a \cdot z_{d,a,r,a'} \geq \frac{\text{RCAP}_{\text{UP}}}{\text{BCY}} \cdot \pi_{a'}^{\text{lw}} \cdot \text{DvB}_{r,a'}, \quad (25)$$

$$\sum_{(d,a):(d,r) \in \mathcal{DR}} \text{DASH}_a \cdot z_{d,a,r,a'} \leq \frac{\text{RCAP}_{\text{UP}}}{\text{BCY}} \cdot \pi_{a'}^{\text{up}} \cdot \text{DvB}_{r,a'}. \quad (26)$$

Path B endogenous-demand block.

(S13) SAF production link.

$$Q_{\text{SAF}} = \sum_{(d,a,r,a')} \text{BCY} \cdot \text{RASH}_{a'} \cdot z_{d,a,r,a'}. \quad (27)$$

(S14) SOS2 simplex.

$$\sum_{k \in K} \lambda_k = 1, \quad \lambda_k \geq 0, \quad (\lambda_0, \dots, \lambda_N) \in \text{SOS2}. \quad (28)$$

(S15) Quantity link.

$$Q_{\text{SAF}} = \sum_{k \in K} q_k \lambda_k. \quad (29)$$

(S16) Consumer-benefit link.

$$I_{\text{CS}} = \sum_{k \in K} c_k \lambda_k. \quad (30)$$

A.5 Objective

The model minimizes (negative welfare):

$$\begin{aligned} \min \quad & \delta \cdot \left[\sum_{(h,d,a)} (\text{ARScost}_h + \text{ARHcost}_h + \text{AtcH2D}_{h,d}) \cdot x_{h,d,a} \right. \\ & + \sum_{(d,a,r,a')} (\text{AtcD2B}_{d,r} + \text{OPEXD} + \text{OPEXB} \cdot \text{BCY} \cdot \text{RASH}_{a'}) \cdot z_{d,a,r,a'} \\ & + \text{CAPEXD} \cdot \sum_{(d,a)} \text{DvD}_{d,a} + \text{CAPEXB} \cdot \sum_{(r,a')} \text{DvB}_{r,a'} \\ & \left. - I_{\text{CS}} \right]. \end{aligned} \quad (31)$$

The discounted single-year welfare $W = \delta (I_{\text{CS}} - C_{\text{undisc}}) = -\text{Obj}$ is reported in the main text. Producer surplus is then $\text{PS} = \delta (P^* Q^* - C_{\text{undisc}})$ and consumer surplus is $\text{CS} = \frac{1}{2} \delta (a - P^*) Q^*$, where $P^* = a - bQ^*$ is the market-clearing price.

A.6 Implementation notes

The model is built in `gurobipy` via direct `m.addVar` / `m.addConstr` construction (no MVar/sparse-tensor wrappers, since the problem-specific sparsity is best exploited by hand). Scenario uncertainty (§3.5) enters by re-computing a_s and the breakpoint vector $c_k^{(s)} = a_s q_k - \frac{1}{2} b q_k^2$ per scenario; the SOS2 simplex (S14) is unchanged. The Quick-Win HN evaluation of §5.6 fixes the binaries via $\text{lb} = \text{ub} = \hat{y}_{d,a}$ and $\text{lb} = \text{ub} = \hat{y}_{r,a'}$, collapsing (S2)–(S6), (S8)–(S12) to LP-only constraints.

B Quick-Win upper bound: derivation and properties

This appendix establishes the formal status of the Quick-Win bound $\mathbb{E}[\text{PVC}_{\text{RP}^*}] \leq \min_c \mathbb{E}[\text{PVC} \mid y_c^*]$ used in §5.5 and discusses its sharpness. We work with the cost-minimization convention throughout: lower PVC is better.

B.1 Notation

Let $S = \{1, \dots, 20\}$ index the KDE scenarios, p_s the scenario probabilities, and $\mathcal{Y}$ the set of feasible non-anticipative first-stage plans (binary placements satisfying §A constraints (S2)–(S6) and (S8)–(S12)). For a scenario s and a plan $y \in \mathcal{Y}$, the LP-tight scenario cost is

$$\text{PVC}(y, s) = \min_{x, z, \text{slack} \geq 0} \{ C_{\text{chain}}(x, z) - \text{PSAF}_s \cdot \text{Production}(z) + \$3.50 \cdot \text{slack} \mid (\text{S1}), (\text{S7}), (\text{S13}_{\text{Path A}}) \},$$

where the demand is enforced as $\text{Production}(z) + \text{slack} \geq Q_{\text{baseline}} = 3 \times 10^9$ (Path A; the slack term is absent in the Path B analysis of §5.6). The recourse-problem optimum is

$$\mathbb{E}[\text{PVC}_{\text{RP}^*}] = \min_{y \in \mathcal{Y}} \mathbb{E}_s[\text{PVC}(y, s)] = \min_{y \in \mathcal{Y}} \sum_s p_s \cdot \text{PVC}(y, s).$$

The wait-and-see expected cost is $\mathbb{E}[\text{PVC}_{\text{WS}}] = \sum_s p_s \cdot \min_y \text{PVC}(y, s)$.

B.2 The Quick-Win upper bound

Let $y_1^*, \dots, y_{20}^* \in \mathcal{Y}$ be any finite collection of non-anticipative plans (in our application, the 20 wait-and-see solutions, but the bound below holds for any feasible candidate set).

Proposition (Quick-Win upper bound). *For any finite candidate set $\{y_1^*, \dots, y_C^*\} \subseteq \mathcal{Y}$,*

$$\mathbb{E}[\text{PVC}_{\text{RP}^*}] \leq \min_{c \in \{1, \dots, C\}} \mathbb{E}[\text{PVC} \mid y_c^*].$$

Proof. Each $y_c^* \in \mathcal{Y}$ is feasible for the recourse minimization, so $\mathbb{E}[\text{PVC}_{\text{RP}^*}] = \min_{y \in \mathcal{Y}} \mathbb{E}_s[\text{PVC}(y, s)] \leq \mathbb{E}_s[\text{PVC}(y_c^*, s)]$ for every c . Taking the minimum over c on the right-hand side preserves the inequality. $\square$

Remark (Sharpness). The bound is tight if and only if some $y_{c^*}^*$ achieves the recourse-problem optimum. In the wait-and-see candidate construction, this is rare: each y_c^* is optimal for scenario c only, and there is no a priori reason for any single scenario’s wait-and-see plan to also be the recourse optimum. The bound is therefore strictly an upper bound; tightening it requires either enriching the candidate set (e.g., with combinations of wait-and-see plans, or expected-value-of-information solutions) or solving the recourse problem directly.

B.3 Why the corresponding lower bound is not reported

The expected-value-of-perfect-information non-negativity inequality $\mathbb{E}[\text{PVC}_{\text{WS}}] \leq \mathbb{E}[\text{PVC}_{\text{RP}^*}]$ holds in theory: scenario information cannot harm the planner. In our numerical setup, however, the wait-and-see cost

$$\mathbb{E}[\widehat{\text{PVC}}_{\text{WS}}]_{\text{reported}} = \sum_s p_s \cdot \widehat{\text{PVC}}_{\text{WS}_s}$$

is constructed from MIP incumbents at a 5% optimality-gap target, so $\widehat{\text{PVC}}_{\text{WS}_s} \geq \text{PVC}_{\text{WS}}^{\text{true}}(s)$ for each s , with the gap unbounded a priori. In our data, $\mathbb{E}[\widehat{\text{PVC}}_{\text{WS}}] = \1.541 B/yr exceeds $\min_c \mathbb{E}[\text{PVC} \mid y_c^*] = \0.925 B/yr , which is incompatible with the standard inequality. We interpret the gap as $\mathbb{E}[\widehat{\text{PVC}}_{\text{WS}}] \geq \mathbb{E}[\text{PVC}_{\text{WS}}]^{\text{true}} \geq \0.925 B/yr is not implied by the data; the reported value carries an upward bias of at least \$616 M/yr if the standard SP inequality is to hold. Re-solving the wait-and-see sweep at a tighter gap target ($\leq 0.5\%$, an estimated 100–200 CPU-hour task) would resolve this; we have not undertaken it for the present paper, and accordingly do not report a tight EVPI estimate. The Quick-Win *upper* bound on $\mathbb{E}[\text{PVC}_{\text{RP}^*}]$ is unaffected by this issue and stands as a clean methodological deliverable.

B.4 Implementation and computational efficiency

The 20 LP solves per candidate are accelerated by primal-simplex warm-start: all 400 LPs share a single LP build, with the per-scenario change being a parametric rewrite of the revenue coefficient on $z[\cdot]$ (specifically $\text{Obj}_z = -\text{PSAF}_s \cdot \text{BCY} \cdot \text{RASH}_{a'} \cdot \delta + (\text{non-PSAF cost})$). Each subsequent LP solves in roughly 0.8 s after the first. Total wall time was 18.8 minutes for the 400-LP sweep — compared to a few hours to a day estimated for an integer Benders convergence at our master scale (§5.5), and unbounded for the failed extensive form. The implementation is in `python_model/quick_win_rp_lb.py`; the 20×20 expected-cost matrix is at `python_model/stage2_sp_out/quick_win_pvc_matrix.csv`.

C Calibration code listing

The full Path B demand-curve calibration is in `python_model/path_b_demand_calibration.py`; it produces `python_model/stage2_out/path_b_calibration.csv` with 11-breakpoint SOS2 inputs for $\varepsilon \in \{-0.3, -0.5, -1.0\}$.